Python Interview & Coding Questions

Fresher / AI-ML / Python Developer Preparation

1. Python Basics

1. What is Python? What are its main features?

2. What is the difference between Python 2 and Python 3?

3. What are variables in Python?

4. What are Python's built-in data types?

5. What is dynamic typing?

6. What is indentation and why is it important in Python?

7. What is the difference between == and is?

8. What are mutable and immutable objects?

9. What is type casting? Give examples.

10. What is None in Python?

2. Data Structures

1. What is the difference between a list and a tuple?

2. What is the difference between a set and a dictionary?

3. How do you remove duplicates from a list?

4. How do you find the maximum and minimum value in a list?

5. How do you reverse a list?

6. What is list slicing?

7. What is dictionary comprehension?

8. What is a set comprehension?

9. How do you sort a list in Python?

10. How do you iterate through dictionary keys and values?

3. Functions

1. How do you define a function in Python?

2. What are positional and keyword arguments?

3. What are *args and **kwargs?

4. What is a default argument?

5. What is a lambda function?

6. What is recursion?

7. What is the difference between return and print?

8. What is variable scope in Python?

9. What are local and global variables?

10. What is a higher-order function?

4. OOP in Python

1. What is object-oriented programming?

2. What are classes and objects?

3. What is the __init__() method?

4. What is inheritance?

5. What is polymorphism?

6. What is encapsulation?

7. What is abstraction?

8. What is method overriding?

9. What is the difference between instance, class, and static methods?

10. What is self in Python?

5. Exception Handling & Files

1. What is exception handling?

2. Explain try, except, else, and finally.

3. What is the difference between an error and an exception?

4. How do you raise a custom exception?

5. How do you open and close a file in Python?

6. What are common file modes such as r, w, a, and x?

7. Why is the with statement useful when working with files?

6. Advanced Python

1. What is a generator?

2. What is the difference between yield and return?

3. What is an iterator?

4. What are decorators?

5. What is a context manager?

6. What is shallow copy vs deep copy?

7. What is a Python module?

8. What is a package?

9. What is pip?

10. What is a virtual environment and why is it useful?

7. Python for Data Science / AI-ML

1. Why is Python widely used in Data Science and AI/ML?

2. What is NumPy and why is it used?

3. What is Pandas and why is it used?

4. What is the difference between a Series and a DataFrame?

5. How do you handle missing values using Pandas?

6. What is the difference between loc and iloc?

7. What is scikit-learn?

8. How do you split data into training and testing sets?

9. What is the purpose of feature scaling?

10. How can Python be used to build an ML model from data preprocessing to prediction?

8. Coding Questions

1. Write a Python program to reverse a string.

2. Write a Python program to check whether a string is a palindrome.

3. Write a Python program to find the factorial of a number.

4. Write a Python program to generate Fibonacci numbers.

5. Write a Python program to check whether a number is prime.

6. Write a Python program to count the frequency of characters in a string.

7. Write a Python program to find duplicate elements in a list.

8. Write a Python program to find the second-largest number in a list.

9. Write a Python program to remove duplicate elements from a list.

10. Write a Python program to find common elements between two lists.

11. Write a Python program to count vowels and consonants in a string.

12. Write a Python program to check whether two strings are anagrams.

13. Write a Python program to calculate the sum of digits of a number.

14. Write a Python program to find the missing number in a sequence.

15. Write a Python program to sort a list without using the built-in sort() method.

9. Fresher Interview Questions

1. Explain one Python project you have built.

2. Why did you choose Python for your project?

3. Which Python libraries did you use and why?

4. How did you debug errors in your project?

5. How did you handle missing or incorrect data?

6. How did you save and load a trained ML model?

7. How would you expose a Python ML model through an API?

8. What is the difference between Flask, FastAPI, and Streamlit?

9. How do you manage Python project dependencies?

10. What are the steps you follow when starting a new Python project?

Quick Preparation Checklist

• Practice Python syntax, data types, loops, conditions, functions, and OOP.

• Solve at least 15–20 coding problems without copying solutions.

• Revise lists, tuples, sets, dictionaries, comprehensions, and string operations.

• Be ready to explain your projects clearly, including your role and technologies used.

• For AI/ML roles, revise NumPy, Pandas, scikit-learn, preprocessing, model evaluation, and basic APIs.

• Practice explaining your code aloud as if you are in an interview.